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Regular Article

Near-infrared and pH-responsive carbon dots/bergenin for biological imaging and chemo-photothermal synergistic tumor therapy

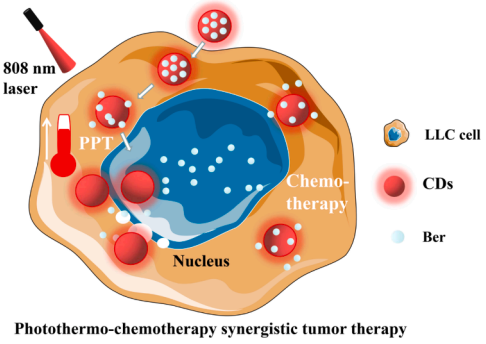
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HIGHLIGHTS

- CDs exhibited long-term photostability and remarkable photoconversion efficiency. External NIR light was applied to convert light energy into heat energy via non-radiative transitions.
- CDs acted as a drug carrier, photothermal agent and fluorescence probe in the chemo-photothermal synergistic therapy.
- CDs/Ber have great potential in dual-triggered drug delivery and thermo-chemotherapy, and acted as an excellent candidate for efficient cancer therapy.
- The CDs/Ber system could mediate synergistic chemotherapy and PTT to effectively eradicate tumors through a single-dose administration and single NIR irradiation.

GRAPHICAL ABSTRACT



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ABSTRACT

Photothermo-chemotherapy has shown a synergistic anti-cancer efficiency and can enhance the therapeutic effect of simple chemotherapy. The photothermal conversion characteristics of carriers are vital in photothermal chemotherapy. In this study, we developed functional carbon dots (CDs) as a photothermal agent and carrier to load bergenin (Ber) for simultaneous biological imaging and photo-chemothermal synergistic therapy. CDs were successfully synthesized using a one-step solvothermal method with citric acid and polyethyleneimine in formamide. Surface morphology, chemical composition and structure, luminescent properties, and photothermal properties of synthesized CDs and CDs/Ber were investigated. The CDs (<10 nm) displayed a mono-dispersed state and uniform spherical morphology with a graphite-like structure. Ber was loaded onto CDs by π-π stacking and intermolecular hydrogen bonds, with a high drug loading efficiency of 95.8 %. *In vitro* release studies verified pH-dependent sustained drug release, with release rates of 37.84 ± 1.67 % (pH 6.5) and 29.19 ± 0.28 %

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(pH 7.4), which conformed to the Weibull model. The CDs/Ber had excellent photothermal performance, stability, and high conversion efficiency (54.9 %) by non-radiative transitions. The CDs acted as an ideal theranostic and photothermal agent for fluorescence imaging, photothermal imaging, ablation, and necrosis of cancer cells. CDs/Ber exhibited excellent water-solubility, biocompatibility, and passive tumor-targeting capability. Cytotoxicity experiments proved that combined therapy of CDs/Ber induced the highest cell-killing level. The biodistribution and pharmacodynamics experiments of CDs/Ber in the lung tumor model within 14 days post-exposure to one-time near-infrared radiation (NIR) irradiation indicated that CDs prolonged the residence time of Ber in tumor tissues and actualized effective synergistic photothermal chemotherapy, yielding a true "1 + 1 > 2" effect. The results of this study indicated that CDs/Ber had great potential in dual-triggered drug delivery and photothermal-chemotherapy, and acted as an excellent candidate for efficient cancer therapy.

1. Introduction

According to the International Agency for Research on Cancer, the incidence rate (12.4 %) and mortality rate (18.7 %) of lung cancer ranked first in 2022 [1], while the report of the National Cancer Center of China showed that in 2022, there would be an increase of 1.06 million cases of lung cancer, with a death toll of 730,000, making it the cancer with the highest incidence rate and mortality rate [2]. In terms of pathology, lung cancer can be divided into small-cell lung cancer (SCLC) and non-small-cell lung cancer (NSCLC). NSCLC accounts for 80–85 % of all lung cancers, with 75 % of patients being diagnosed at an advanced stage [3], which can lead to patients missing the optimal treatment window, resulting in a 5-year survival rate of only 5.5 % for NSCLC patients [4]. Currently, chemotherapy is one of the main treatments for NSCLC, but the survival rate of patients with advanced lung cancer who receive traditional chemotherapy is very low. Adverse reactions such as myelosuppression, liver and kidney function damage, and hair loss often occur, seriously affecting the life quality of patients. Therefore, it is particularly important to improve treatment efficacy and patient survival rate and reduce adverse reactions.

Traditional Chinese medicine (TCM), as a disease treatment approach in China, has many advantages in cancer treatment, such as fewer side effects, non-invasive, low cost, and benefiting a wide range of people. In recent years, many studies have shown that various components of TCM have anti-cancer activity [5,6]. Compared with traditional chemotherapeutic drugs, natural products have gradually become attractive therapeutic agents due to their less cytotoxic and side effects. Bergenin (Ber) is an herbal bioactive compound obtained from *Bergenia purpurascens* (Hook. f. & Thomson) Engl. and *Ardisia crenata* Sims. In recent years, people have discovered that Ber not only has anti-inflammatory, liver-protective, and immune-enhancing effects [7] but also exhibits anti-cancer activity, such as against colorectal cancer [8], lung cancer [9], and cervical cancer [10]. Studies have shown that Ber has a promising anti-tumor effect against NSCLC by targeting survivin, making it a potential therapeutic drug for NSCLC [11]. However, it has low solubility and low permeability and is easily eliminated by the P-glycoprotein (P-gp) efflux pump, resulting in poor bioavailability [12]. Therefore, developing tumor-targeted drug delivery systems (DDSs) is an effective way to overcome these limitations [13]. In light of the multi-level complexities and variability of tumors, it is necessary to develop DDSs that combine diagnosis, chemotherapy, and other treatments to improve the effectiveness of treatment and reduce side effects.

As a new emerging physical therapy, photothermal therapy (PTT) is becoming a very promising cancer treatment approach due to its non-invasiveness, low cytotoxicity, absence of drug resistance, and few side effects [14]. PTT typically employs near-infrared radiation (NIR) light-absorbing agents that convert laser energy into heat to kill cancer cells, generating irreversible cellular damage and tumor destruction [15]. Compared to chemotherapy, the therapeutic effects of PTT occur at the tumor sites, resulting in low side effects, high specificity, and low invasiveness. Efficient PTT agents should exhibit strong absorbance in the NIR region, high photoconversion efficiency, and high systemic safety. Currently, PTT agents have been derived from precious metal-based nanomaterials [16] and organic NIR chromophores [17]. These agents typically exhibit strong absorption in the NIR region and relatively high photoconversion efficiency. However, precious-metal-based nanomaterials have the potential for long-term toxicity, non-biodegradability, and high cost, which severely limit their application in the body. The low photostability and poor hydrophilicity of traditional NIR dyes also restrict their therapeutic applications. Additionally, their poor optical stability can lead to the deformation of nanoparticles and reduced photothermal efficiency during continuous irradiation [18]. Therefore, it is very critical for the development of DDSs with good biocompatibility, stable biological imaging, and a high photothermal conversion rate.

With the advancement of medical nanotechnology, novel nanoparticles have become popular materials for the diagnosis and treatment of a variety of cancers [19,20]. Drug delivery materials with multifunctional photothermal properties have attracted considerable attention in recent years, owing to their potential for spatiotemporally controlled therapy and synergistic diagnostic-therapeutic integration [21,22]. Carbon dots (CDs) have become potential nanomaterials due to their fluorescence monitoring capability, high photostability, water dispersibility, high photoconversion efficiency, excellent biocompatibility, facile preparation and modification [23]. The tunable bandgap of CDs enables NIR absorption and emission, facilitating deep tissue penetration. Therefore, CDs can serve as potential photothermal agents in cancer treatment [24,25]. Since Ber is an aromatic compound, CDs can act as effective carriers for the delivery of Ber through π-π stacking interactions [26]. Additionally, CDs can also effectively improve the solubility of Ber [27]. In this research, we fabricated a nano-platform that had simultaneous multifunctional applications for photothermal therapy, chemotherapy, and biological imaging by loading Ber with CDs. As elucidated in Scheme 1, CDs, as a drug carrier of Ber, could achieve synergistic anticancer efficacy by absorbing NIR. The morphology, size, chemical composition and structure, and optical properties of CDs/Ber were researched. Additionally, photothermal efficiency, photothermal conversion capability, fluorescence imaging, drug loading, and release profiles of CDs/Ber were investigated. Furthermore, the *in vitro* and *in vivo* combined antitumor therapies were evaluated to demonstrate synergistic chemo-photothermal therapy.

2. Materials and methods

2.1. Reagents

All these chemicals used were of analytical grade and were used as received without further purification. Citric acid (CA) and formamide were acquired from Sinopharm Chemical Reagent Co., LTD. Polyethyleneimine (PEI, average Mw ~ 800 by LS, average Mn ~ 600 by GPC, 99 %) was obtained from Shanghai Macklin Biochemical Technology Co., LTD. Bergenin (Ber) was obtained from Chengdu Push Biotechnology Co., LTD. DMSO, penicillin–streptomycin, and trypsin-EDTA were bought from Suolaibao Bio-Technology Co., LTD. Roswell Park Memorial Institute (RPMI-1640) medium was purchased from Thermo Fisher Scientific Inc. Fetal bovine serum (FBS) was purchased from Hangzhou Siqijing Biological Co., LTD. Cell Counting Kit-8 (CCK-8) and Live & Dead™ Animal Cell Viability/Toxicity Assay Kit were acquired

from UElandy (Jiangsu China). Hoechst 33,342 was purchased from Wuhan Saiweier Biotechnology Co., LTD.

2.2. Synthesis of CDs

The CDs were synthesized using a solvothermal synthesis method [28]. Briefly, 10 g of CA and 6.3 g of PEI were dissolved in formamide (200 mL) under stirring to form a homogeneous solution. The above solution was transferred to a Teflon autoclave and heated at 180 °C for 4 h. The product was cooled and centrifuged at 8000·min^{-1} for 15 min, and then filtered with a 0.22 μm membrane filter. The final supernatant solution was collected and dialyzed using a dialysis bag (MWCO 500 Da). The final product was collected by lyophilization.

2.3. Characterization

Transmission electron microscopy (TEM) and high-resolution TEM (HR-TEM) images were taken on a Talos F200S G2 S/TEM microscope. Fourier transform infrared (FT-IR) spectra were recorded on an IRTracer-100 FT-IR spectrometer. X-ray photoelectron spectroscopy (XPS) was carried out with NexsaX X-ray electron spectroscopy. X-ray diffractometer images (XRD) were recorded on a SmartlabX X-ray diffractometer. Raman spectrum was obtained on the Renishaw in Via-Reflex spectrometer at the excitation wavelength of 532 nm. UV–vis absorption spectra were performed on a TU-1810 UV–vis spectrophotometer. Fluorescence emission and excitation spectra were recorded on a F98 spectrophotometer. Zeta potentials and dynamic light scattering (DLS) were measured on a Delsa Maxpro Beckman particle size analyzer. The fluorescence lifetime decay curve was recorded on an FLS 980 static/transient fluorescence spectrometer. The temperature change and photothermal images were performed by an HM-TPH21Pro-3AQF handheld thermal imaging temperature meter. Cellular images were performed by confocal laser scanning microscope (CLSM, Zeiss LSM 800). CCK-8 results were collected by a Multiskan Sky microplate absorbance reader.

2.4. Photothermal efficiency in vitro

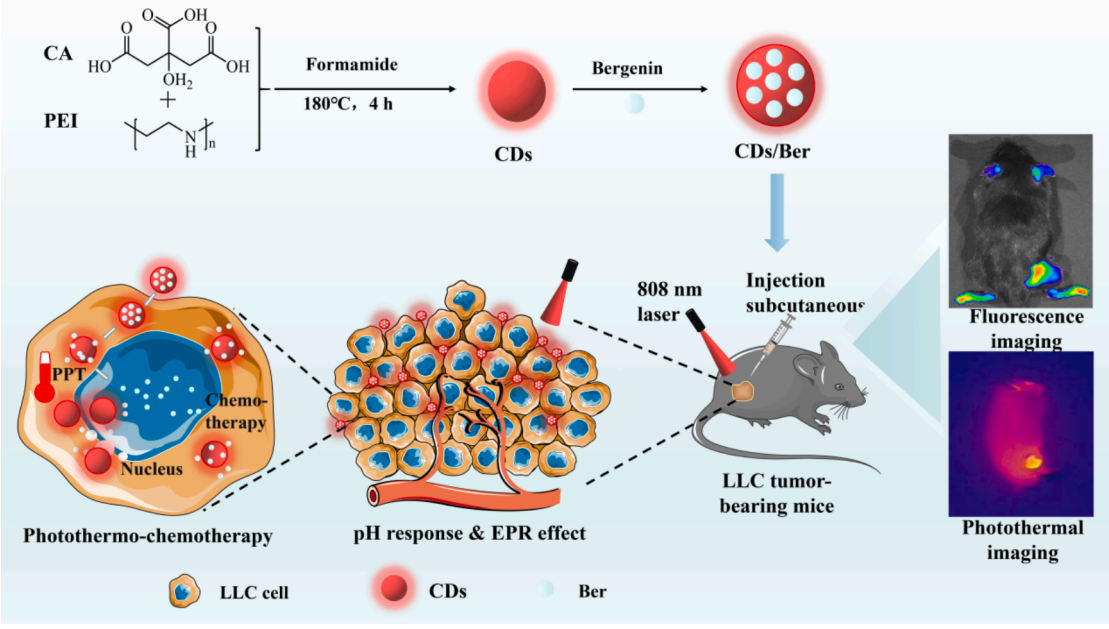
The photothermal efficiency of CDs and CDs/Ber was measured with a DT1310 temperature sensor. The CDs and CDs/Ber of various concentrations (0, 25, 50, 100, 200 and 400 μg·mL^{-1}) were irradiated with 808 nm laser at a power density of 2 W·cm^{-2} for 5 min to measure the photothermal performance. Pure water was used as a control. Furthermore, samples (400 μg·mL^{-1}) were exposed to a NIR laser (808 nm) with different laser power densities (0, 0.5, 1, 1.5, and 2 W·cm^{-2}), and the temperature elevation (ΔT) was recorded by a thermal imaging temperature meter. CDs dispersions (400 μg·mL^{-1}) were irradiated for 5 min at a power density of 2 W·cm^{-2}, followed by natural cooling to room temperature without irradiation, which was repeated 5 times to evaluate the photothermal stability. The determination of photoconversion efficiency (η) was referred to in the previous report [29].

2.5. Drug loading efficiency and drug release

Ber was employed as the model drug to investigate the drug loading capacity and controlled release profiles of the drug delivery system. A 1 mg·mL^{-1} solution of Ber was prepared using methanol, mixed with 1 mg·mL^{-1} CDs in a 1:1 ratio. Subsequently, the mixture was continuously stirred in the dark at room temperature for 24 h. After incubation, the mixture was dialyzed using a dialysis bag (MW 500 Da) in the dark. The absorbance readings of the free Ber were determined at 275 nm using UV–visible spectroscopy. The drug loading efficacy of Ber was estimated by the previous report [30]. The drug release behavior from the CDs/Ber complex was assessed at pH 6.5 and pH 7.4 PBS by *in vitro* method. Briefly, the prepared CDs/Ber complex was suspended in a release medium at 37 °C and continuously stirred at 60 rpm. At regular intervals, 2 mL samples were withdrawn and replaced with an equivalent volume of fresh PBS buffer. The samples were withdrawn until the saturation point of the drug release. Then, the amount of released Ber was analyzed using a UV–vis spectrophotometer at 275 nm at each time point. The cumulative drug release was calculated, and drug release patterns at different pH were compared and correlated.

2.6. Cytocompatibility analysis and cell imaging

The L929 Mouse Fibroblasts (L929 cells) and HPDE6-C7 (H6C7) cells were seeded at a density of 5 × 10^3 cells/well in a 96 well-plate and cultured in RPMI-1640 medium at the standard conditions of 5 % CO_2 at 37℃. After culturing for 24 h, cells were incubated with 100 μL of RPMI-1640 containing different concentrations of CDs and CDs/Ber for 24 h.



Scheme 1. Schematic diagram of the antitumor mechanism of *in vivo* photothermal-chemo synergistic therapy for cancer.

Then, 50 μL of the 10 % CCK-8 solution was added, and the mixture was further incubated for 4 h. The optical density of the samples was analyzed using a microplate reader at 450 nm, and cell viability was calculated. LLC cells were seeded at a density of 2.5×10^6 cells/well onto 24-well plate in LLC cell-specific culture medium at 37 ℃ for 24 h. The cells were incubated with CDs, and CDs/Ber for 1, 2, 4, and 8h. Then, the cell monolayer was washed 3 times with PBS (pH 7.4) and fixed with 4 % paraformaldehyde for 15 min. Finally, Hoechst 33,342 staining solution was added to each well and the incubation was continued for 15 min. Cellular uptake and imaging were detected by a confocal laser scanning microscope and an inverted fluorescence microscope.

2.7. Cellular photothermal effect induced by NIR

The LLC cells and A549 cells were seeded in a 96-well plate and incubated at 37 ℃ with 5 % CO_2 for 24 h for 4 h. Cells were treated with different concentrations of Ber, CDs, and CDs/Ber, and then incubated for 4 h. Routinely, after cell digestion, collection and resuspension, samples were irradiated with or without NIR laser of 2.0 W·cm^{-2} for 5 min. The standard CCK-8 assay was employed to measure cell viability. The calculation of the combination index (CI) is followed by the method: $CI = [IC_{50(\text{combined therapy})}/IC_{50(\text{chemotherapy alone})}] + [IC_{50(\text{combined therapy})}/IC_{50(\text{PTT alone})}]$. $CI < 1$ shows synergism; $CI > 1$ means antagonism; $CI = 1$ represents additive effect [31,32]. Finally, the live and dead cells were distinguished by PI and calcein AM and observed under an inverted fluorescence microscope.

2.8. In vivo biodistribution

In vivo, biodistribution imaging of CDs/Ber was performed in 6- to 8-week-old C57BL/6J female mice. The mice (18–22 g) were used to establish the LLC tumor model by subcutaneous injection of LLC tumor cells (2×10^6 cells in 150 uL of PBS) into the right rear flank. When the tumor size reached approximately 100 mm^3, LLC tumor-bearing mice were injected separately with 100 μL of CDs/Ber (200 μg·mL^{-1}). The autofluorescence was removed by the spectrum unmixing software. Mice were euthanized and sacrificed at the given times (1, 2, 4, 8, 12, 24 36, and 48 h). Then, the tumors and major organs (heart, liver, spleen, lung, kidney, and tumors) were collected and imaged by a small animal imaging system.

2.9. In vivo combination therapy and histological examination

Tumor-bearing mice were randomly divided into four groups (five mice per group) and separately received treatment with normal saline (control), CDs, Ber, and CDs/Ber. Each group injected 100 uL of the saline, CDs, Ber, and CDs/Ber (200 μg·mL^{-1}) into the tumor before 4 h, and then the 808 nm laser irradiation was carried out on tumor regions for 2 min at a power density of 0.8 W·cm^{-2}. During the irradiation period, images of mice and the temperatures of tumor sites were recorded by a thermal imaging temperature meter. The body weights and tumor volumes of these mice were measured every other day. The tumor volume was determined according to the following equation: $V = (A \times B^2)/2$, where A and B are the maximum and minimum diameter of the tumor, respectively. After 14 days of treatment, the various organs (heart, liver, spleen, lung, and kidney) and tumor tissues were collected, fixed with 4 % paraformaldehyde solution, and stained with H&E for further histological examination.

2.10. Statistical analysis

All experiments were performed at least 3 times and presented as mean ± SD. Statistical significance of the data was analyzed using the two-tailed Student's *t*-test: $^*p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$, $^{****}p < 0.0001$. Multiple groups of data were analyzed by One way of ANOVA to evaluate the statistical significance ($^*p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$, $^{****}p < 0.0001$).

3. Results and discussion

3.1. Preparation and characterization of CDs and CDs/Ber

CDs were facilely synthesized through a solvothermal heating and formamide solution that included citric acid and polyethyleneimine at 180 ℃ for 4 h in a high-pressure reactor, followed by centrifugation and dialysis purification (Scheme 1). The morphology and size distribution of the synthesized CDs and CDs/Ber were first characterized by TEM (Fig. 1A–H). As shown in Fig. 1A, the CDs were well-dispersed and approximately spherical with a narrow size distribution in the range of 1.25–2.93 nm. The DLS analysis showed the hydrodynamic size of 2.68 ± 0.49 nm in line with the particle size observation from the TEM images and corroborating the considerable water dispersibility of the CDs (Fig. 1B). The HR-TEM image showed that CDs exhibited lattice fringes of 0.21 nm (Fig. 1C), which were located in the graphite (1 0 0) lattice plane [33]. The surface of CDs featured a distinct honeycomb-like structure in the enlarged HR-TEM image (Fig. 1D). CDs/Ber appeared as highly dispersed near-spherical nanoparticles (Fig. 1E) and its average size was determined by DLS to be 7.33 ± 2.22 nm (Fig. 1F). The size distribution confirmed an increase in the size of CDs/Ber and the successful construction of CDs/Ber assemblies. The CDs/Ber possessed clear lattice stripes, a lattice spacing, and a honeycomb structure (Fig. 1G–H), similar to CDs. These results indicated that Ber had successfully been loaded onto the CDs, increasing their particle size, but without affecting their structure [34]. Additionally, the surface charge of the CDs was −14.68 mV, which could be attributed to the presence of oxygen- and nitrogen-based functional groups such as hydroxyl, epoxy, carbonyl, and amino groups [35]. The loading of Ber reduced the negative charge (−11.92 mV) of CDs/Ber. This result indicated that the loading of Ber might have formed a shielding effect on CDs, thereby weakening the electronegativity of CDs.

To obtain the structural information about CDs, XRD patterns were recorded and shown in Fig. 1I. A diffraction peak of CDs was observed at 26.6°, indicating a graphite-like structure (002). Raman spectroscopy was a useful tool for the structural characterization of the CDs. Two characteristic G and D bands of CDs were observed at 1566 and 1361 cm^{-1}, respectively (Fig. 1J). The G band was caused by the plane vibrations of sp^2 hybridized carbon atoms and signified graphitization of the carbon materials. On the contrary, the D band was associated with topological defects or sp^3 hybridized carbon atoms, representing defects and disordered carbon materials [36]. The ordered G band was stronger than the disordered D band, with an intensity ratio (I_D/I_G) of 0.81, which indicated a high degree of graphitization of the carbon nanomaterials, reflecting the good crystalline quality of the CDs [37]. Furthermore, CDs and CDs/Ber were characterized with FTIR as shown in Fig. 1K. The characteristic peaks of CDs at 1084, 1349, 1503 and 1634 cm^{-1} were ascribed to the stretching vibrations of C–O, C–N, C=C and C=O/C=N, respectively. Additionally, the broader peak at 3210 cm^{-1} was attributed to the amino (–NH_2) and hydroxyl (–OH) functional groups [38], indicating that the CDs exhibited good dispersion and strong adsorption [39]. The peaks at 1461, 1512, and 1611 cm^{-1} were characteristic of the benzene ring structure in Ber [40]. For CDs and CDs/Ber, the peak at 3210 cm^{-1} was red-shifted to the peak at 3270 cm^{-1} (the stretching vibrations of the O–H and N–H bonds), which could be attributed to the introduction of Ber and the formation of hydrogen bonds. The characteristic peak of the benzene ring in CDs/Ber was red-shifted to 1647 cm^{-1}, indicating a successful loading of Ber. CDs, which possessed multiple hydrophilic functional groups (hydroxyl, carboxyl, and amino groups), enhanced the solubility of Ber by161-foldwhen loaded onto the CDs, demonstrating their capacity to overcome the inherent hydrophobicity of Ber through supramolecular interactions.

XPS results of CDs further verified the FT-IR analysis. CDs mainly

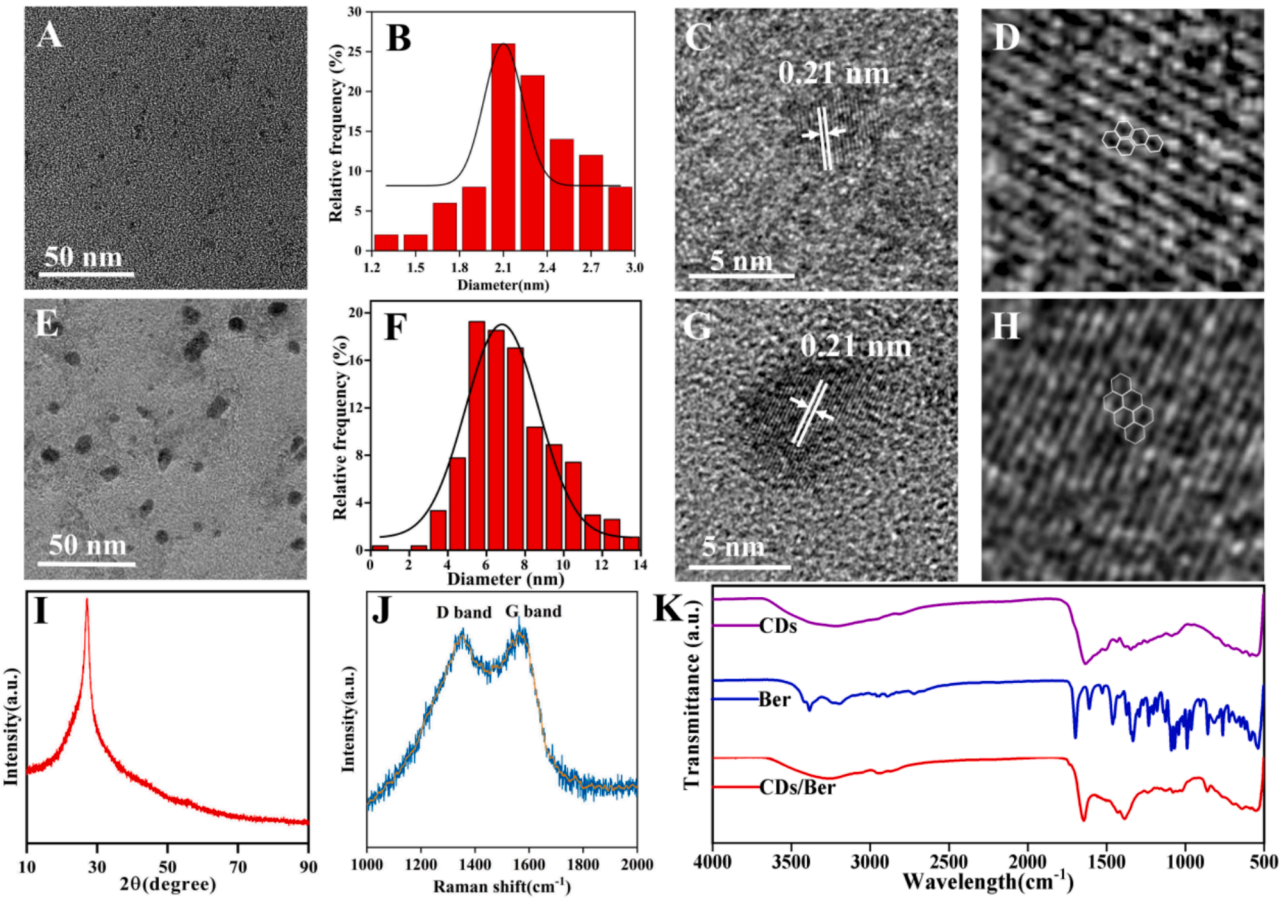


Fig. 1. (A) TEM images, (B) particle size, (C) HRTEM images, and (D) HRTEM magnification image of CDs. (E) TEM images, (F) particle size, (G) HRTEM images, and (H) HRTEM magnification image of CDs/Ber. (I) XRD patterns, (J) Raman spectra, and (K) FT-IR spectrum of CDs.

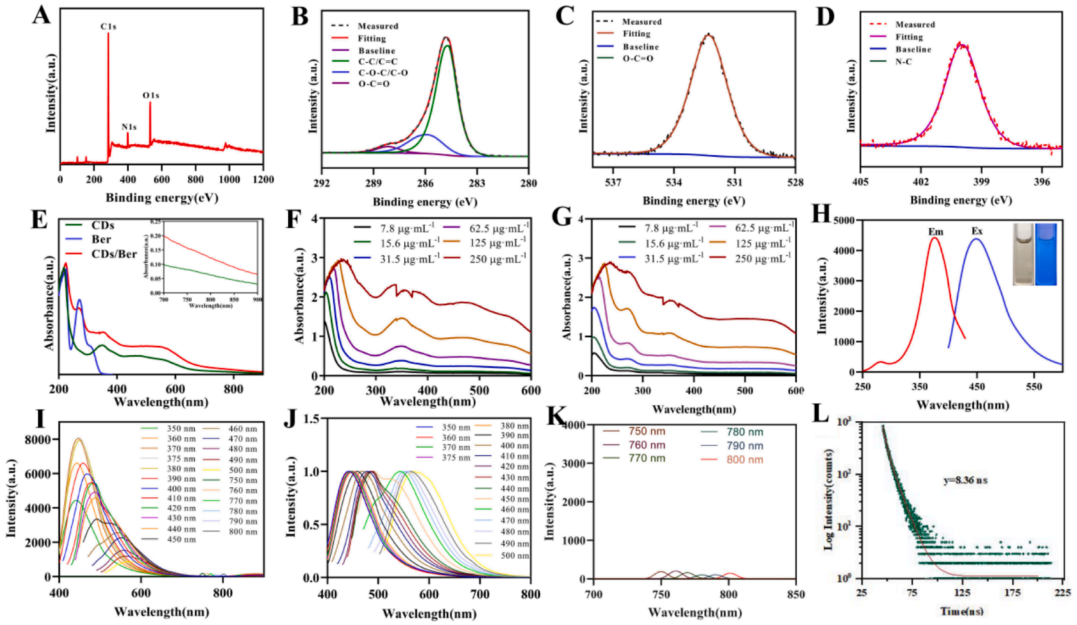


Fig. 2. (A) The full XPS spectrum of CDs. High-resolution (B) C 1s, (C) O 1s, and (D) N 1s XPS spectra of CDs. (E) UV–vis spectra and the absorbance of the CDs/Ber in the NIR region (inset). UV–vis spectra of CDs (F) and CDs/Ber (G) with different concentrations. (H) Fluorescence spectra of CDs (insert: the photos of the aqueous solution of CDs under visible and 365 nm UV illumination). (I) Excitation-dependent fluorescence spectra and (J) normalized fluorescence spectra of CDs. (K) Fluorescence spectra of CDs in the NIR region. (L) Fluorescence decay curves of CDs.

consisted of carbon (80.6 %), nitrogen (7 %), and oxygen (12.4 %) with the binding energy peaks at 284.08, 400.08, and 532.08 eV, corresponding to C 1s, N 1s and O 1s, respectively (Fig. 2A). The result showed that most of the carbon atoms in the citric acid precursor were introduced with or converted to oxygen and nitrogen-containing functional groups. Fig. 2B–D presented the corresponding high-resolution XPS spectra of C1s, O1s and N1s, respectively. The XPS spectra of C1s presented three peaks at 284.7, 285.9, and 288.2 eV (Fig. 2B), corresponding to the C–C/C=C, C–O–C/C–O, and O–C=O bonds, respectively [41]. The O1s XPS spectrum contained a fitting peak at 532.3 eV (Fig. 2C), which could be ascribed to the O–C=O bonds. As shown in the high-resolution N 1s spectrum (Fig. 2D), the deconvoluted peaks represented the presence of N–H bonds (399.9 eV). The multiple groups on the surface of CDs played a crucial role in their drug-loading process [42], and also endowed them with excellent fluorescent properties [43].

Subsequently, the optical properties of CDs were investigated to determine whether CDs were suitable for biological labeling and imaging. The optical properties were measured by UV–vis and photoluminescence (PL) spectra. Fig. 2E showed the CDs possessed a broad absorption range (200–900 nm) between the UV–vis and NIR regions. Two characteristic absorption peaks were at 225 and 354 nm in the UV–vis region (Fig. 2E), and probably corresponding to π-π* transition of C=C at the aromatic region and n-π* transition of the C=O and C=N bonds, respectively [44]. The absorbance of the CDs in the NIR region (inset in Fig. 2E) was similar to those of various graphene-based nanomaterials [45] and carbon dots [46,47] that had been used for cancer photothermal therapy at 808 nm laser irradiation. The CDs also emitted a brilliant blue luminescence upon exposure to a 365 nm UV lamp operated in the dark (inset in Fig. 2H). Moreover, the UV–vis spectrum of CDs/Ber exhibited the same absorption peak at 275 nm as Ber, indicating that Ber had been successfully loaded onto CDs. As Fig. 2F–G, the UV absorbance of both CDs and CDs/Ber increased in a concentration-dependent manner, and the maximum absorption peak of CDs/Ber underwent a slight redshift. Fig. 2H and Fig. S1A indicated that the optimal excitation and emission wavelengths of CDs were located around 375 nm and 445 nm, respectively. It was observed that the light emission of CDs depended on the excitation wavelength (Fig. 2I and Fig. S1B). Through normalization in Fig. 2J, the emission peak showed an obvious redshift (130 nm) with the increase of excitation wavelength like most CDs [48]. This phenomenon was caused by the n-π* transition of non-bonded electrons of carbonyl and carboxyl groups on the surface of CDs. Research indicated that low emission was observed at excitation wavelengths of 808 nm (Fig. 2K), suggesting that most of the absorbed light by the CDs in those excitation wavelengths was converted to heat via non-radiative transitions [35]. This phenomenon was very crucial for the CDs-mediated cancer photothermal therapy upon 808 nm laser irradiation. When CDs were loaded with Ber, their optimum fluorescence excitation and emission spectra remained unaffected by Ber (Fig. S1E–F). It could be known from Fig. 2L that the fluorescence lifetime of CDs was relatively long, up to 8.75 ns. The influence of various ionic conditions on the fluorescence properties of CDs was determined. As indicated in Fig. S1C and G, K^+, Ca^{2+}, and Na^+ exerted an enhancing effect on the fluorescence of CDs and CDs/Ber, whereas Fe^{2+} and Fe^{3+} had a quenching effect on them. K^+, Ca^{2+}, and Na^+ were present in the body in the form of ions and beneficial for imaging of CDs *in vivo*. The influence of different pH on the fluorescence of CDs was investigated in Fig. S1D. The fluorescence intensity of CDs was higher under acidic and alkaline conditions than under neutral conditions. This characteristic was conducive to maintaining the favorable fluorescence performance of CDs in the tumor environment. Additionally, the pH value's effect on CDs/Ber was consistent with that of CDs (Fig. S1H).

3.2. Photothermal effect of CDs/Ber

As a candidate for photothermal anticancer therapy, the photothermal conversion performance of CDs and CDs/Ber was thoroughly studied. To evaluate the capability of CDs and CDs/Ber to induce cancer cell death by PTT, the *in vitro* photothermal behavior of different CDs concentrations (50, 100, 200, and 400 μg·mL^{-1}) was assessed in Fig. S2A. The CDs (Fig. S2A) and CDs/Ber (Fig. 3A) exhibited a concentration-controlled temperature elevation profile upon exposure to an 808 nm NIR laser irradiation for 5 min. Temperature elevation with 1 mL of CDs and CDs/Ber dispersion (400 μg·mL^{-1}) reached 61.67 ± 0.17 °C and 66.67 ± 0.26 °C, respectively. The comparable temperature elevation observed in both groups (CDs and CDs/Ber) suggests that the incorporation of Ber did not significantly compromise the photothermal performance of carbon dots (CDs). Under identical experimental conditions, Ber samples with equivalent load-bearing capacity demonstrated no statistically significant temperature variation. Contrarily, pure water serving as control showed a temperature increase of 2.83 ± 0.31 °C after laser exposure in 5 min. The concentration-dependent temperature maps of CDs and CDs/Ber in Fig. S2B and Fig. 3B were consistent with the results of the temperature variations. Moreover, the temperature of CDs and CDs/Ber aqueous dispersions (400 μg·mL^{-1}) gradually rose with increasing laser power density from 0.25 to 2.0 W·cm^{-2} (Fig. S2E and Fig. 3E). The maximum temperature increment of

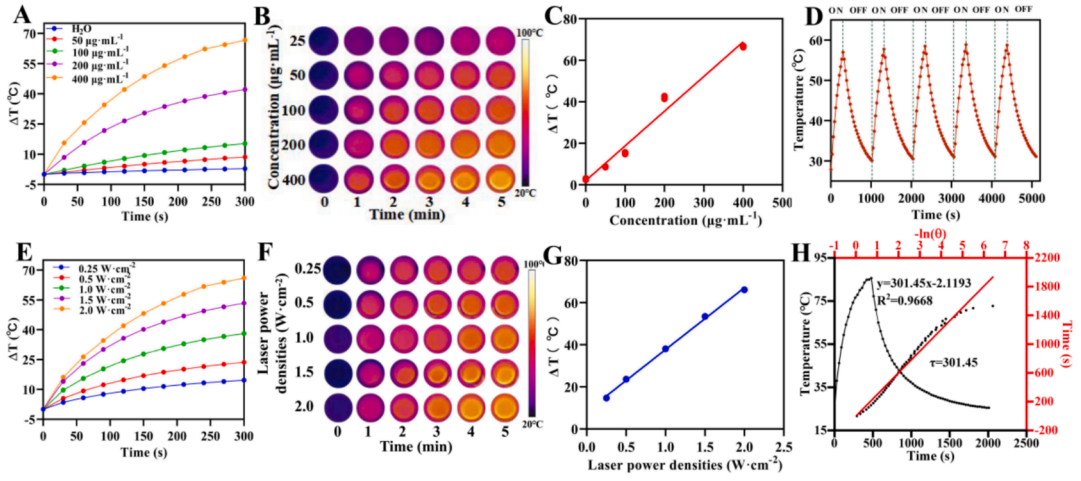


Fig. 3. (A) Temperature variations, (B) photothermal image, and (C) heat generation of CDs/Ber by different concentrations at 5 min upon 808 nm laser irradiation (2 W cm^{-2}) (n = 3). (D) Photothermal stability of CDs/Ber dispersion over 5 laser on/off cycles at 2 W·cm^{-2}. (E) Temperature variations, (F) photothermal image, and (G) heat generation of CDs/Ber by various laser optical power densities at 5 min upon 808 nm laser irradiation. (H) The plot of cooling time data versus −lnθ and the corresponding linear fitting curve.

CDs and CDs/Ber reached 61.63 ± 0.34 and 66.07 ± 0.25 °C after exposure to 808 nm laser irradiation for 5 min at a dose of 2.0 W·cm^{-2}, respectively. Notably, even at a laser power density of 0.5 W·cm^{-2}, the temperature elevation of CDs/Ber reached 23 °C, meeting the requirements for mild antitumor therapy. Therefore, the power density was chosen between 0.5 and 1 W·cm^{-2}, with 0.8 W·cm^{-2} laser dose being more suitable for subsequent photothermal experiments. Photothermal images of CDs/Ber exposed to the 808 nm laser at various power densities were shown in Fig. S2F and Fig. 3F. These results manifested that CDs and CDs/Ber could quickly and efficiently convert the absorbed laser energy into heat. Furthermore, the photostability of the CDs and CDs/Ber was determined via laser on/off cycles (Fig. S2D and Fig. 3D). It was impressive that CDs and CDs/Ber showed good thermal reproducibility and photothermal stability in 5 laser on/off cycles with a laser power density of 2 W·cm^{-2}. This suggested that CDs/Ber could be used to enhance cancer treatment efficacy by repeatedly heating cancer cells *in vivo*. This impressive photothermal stability was a beneficial property already deployed in cancer photothermal therapy with CDs/Ber [49]. On the contrary, gold nanorods as one of the most studied cancer PTT agents, showed a significant decline in photothermal stability after 4 or 5 cycles of laser on/off [50]. The photothermal stability of CDs/Ber was also superior to commercially available indocyanine green dyes. The photothermal ability of the latter rapidly declined during repeated laser on/off cycles [51]. Additionally, the photoconversion efficiency (η) of the CDs and CDs/Ber at 808 nm was calculated to be 54.1 % and 54.91 % according to the obtained data from Fig. S2H, Fig. 3H and Table S1. Additionally, the high conversion efficiency of the CDs/Ber was superior to other CDs from different sources, such as AA-CDs [52], DMCR [53], CDs [54], plys-CDs [55], and ACDs [48] (Table S2). CDs with outstanding photoconversion efficiency and stability could be a promising carrier for PTT.

The heat generation mechanism of carbon-based nanoparticles in PTT was mainly through the stimulation of loosely bound π electrons and subsequent relaxation of excited electrons to generate heat [43]. Research indicated that graphene quantum dots had low fluorescence quantum yield, mainly due to the presence of non-radiative transitions [56]. When carbon-based nanomaterials of low quantum yield absorbed a particular wavelength, they converted most of the absorbed energy into thermal energy instead of light emission. CDs had a low quantum yield of 1.26 % and exhibited weak fluorescence emission at 808 nm laser (Fig. 2K). Therefore, it was reasonably inferred that all the energy absorbed by CDs and CDs/Ber at 808 nm was dissipated as heat through non-radiative transitions. To further explore whether light scattering was involved in enhancing the heat generation of CDs, the effect of different concentrations of the CDs (Fig. S2C) and CDs/Ber (Fig. 3C) and laser power densities (Fig. S2G and Fig. 3G) on the heating performance was measured. If CDs exhibited light scattering, their heat generation curve should deviate from linearity as the concentration increased [35], but the results were all linear. This indicated that the possibility of CDs and CDs/Ber generating heat through light scattering was very low. Therefore, the heat generation mechanism of CDs and CDs/Ber might be non-radiative electron relaxation by internal conversion and vibrational relaxation.

3.3. Drug loading and release studies on CDs/Ber

According to the formula calculation, the average drug loading capacity of CDs/Ber was 95.8 ± 0.54 %, indicating that CDs had a strong drug loading capacity. As an aromatic carbohydrate, Ber with the benzene ring and C=O group exhibited a π-π conjugation effect and formed delocalized large π bonds, which might incline to combine with the CDs plane by π-π stacking interactions [57]. As shown in Fig. 2E, due to the strong π-π stacking interaction between the Ber and CDs, the absorption spectrum of CDs/Ber was subtly red-shifted (~5 nm) [58]. Furthermore, it was hypothesized that drug loading might also involve hydrogen bonds based on the FT-IR result. The pH-dependent release behavior of CDs/Ber was systematically investigated. pH 6.5 was selected as a representative tumor microenvironmental condition based on well-established evidence that the extracellular pH of solid tumors typically ranges between 6.5 and 6.8 due to elevated glycolytic activity and lactic acid accumulation. Fig. 4A presented the drug release curves of CDs/Ber under pH conditions of 7.4 and 6.5. The overall Ber release pattern in different pH media showed biphasic drug release patterns, such as an initial burst release of around 23–33 % of the drug, followed by sustained release over time. The Ber release samples were taken out for 120h, and the maximum Ber release amount was approximately 29–38 % at different pH. Notably, Ber exhibited pH-responsive release kinetics, with significantly accelerated release rates under acidic conditions (37.84 ± 1.67 % at pH 6.5) compared to physiological pH levels (29.19 ± 0.28 % at pH 7.4; $p < 0.01$, Student's *t*-test). The release rates of Ber in weak acidic conditions were higher than those in neutral conditions, which could be ascribed to the protonation of Ber in acidic pH, leading to the attenuation of the interaction between CDs and Ber. This pH-triggered release enhancement aligns with the acidic tumor microenvironment (pH 6.0–6.8), suggesting spatial selectivity for tumor-targeted drug delivery. The drug release studies were also performed at an 808 nm laser to check for any change in the Ber release pattern. There was no significant difference in the Ber release, irrespective of acidic and neutral conditions at laser radiation.

A variety of mathematical models were frequently employed to fit the *in vitro* drug release or dissolution data and to explore the drug release mechanisms (immediate/controlled/sustained release) via linear and nonlinear equations [59]. The cumulative drug release data was fitted using various models, including zero order, first order, Higuchi, Ritger-Peppas, and Weibull. The linear fitting curves of the drug release of CDs/Ber were depicted in Fig. 4B–F. The correlation coefficient (R^2) was the most frequently employed parameter for selecting the appropriate model. The R^2 and other significant values were summarized in Table S3. The Weibull model seemed to optimal fitting model, with the highest R^2 (0.99) at pH 6.5 and 7.4 [60]. Consequently, CDs were capable of preventing premature leakage of Ber during the blood circulation process, circumventing adverse side effects, and enhancing the release of the drug within tumor cells. This was highly beneficial for the effective treatment of cancer.

3.4. Cellular uptake

CLSM was applied to study the uptake of CDs and CDs/Ber by LLC cells as shown in Fig. 5. For the control group, only blue fluorescence stained by Hoechst appeared, and Ber itself would not fluoresce at the excitation wavelengths (Fig. S3). When CDs and CDs/Ber were cultured with LLC cells for 1 h, red fluorescence appeared, and the red and blue fluorescence overlapped, indicating that CDs and CDs/Ber could enter into the nucleus. This phenomenon may be attributed to the small-sized particles of both CDs and CDs/Ber (Fig. 1B and F), which enable their passive diffusion through the nuclear pore complex with a diameter of approximately 9 nm, thereby facilitating nuclear entry. As the incubation time increased, the red fluorescence in the cells became stronger, reaching its maximum at 4 h. Notably, the CDs/Ber group exhibited significantly higher fluorescence intensity compared to the CDs group. This enhancement could be attributed to the Ber-mediated neutralization of the negative surface charges on CDs, which forms near-neutral complexes [61]. The reduced surface charge density minimizes electrostatic repulsion with the anionic cell membrane, thereby promoting cellular uptake and subsequent intracellular accumulation of CDs/Ber, ultimately amplifying the observed fluorescence signal. The observed fluorescence attenuation in both CDs and CDs/Ber groups at 8 h incubation could be attributed to time-dependent cytotoxicity. Prolonged exposure triggers apoptotic/necrotic pathways (Fig. 5), compromising plasma membrane integrity, thereby facilitating nanoparticle efflux into the extracellular milieu or exposing them to degradative enzymes, ultimately quenching fluorescence signals. The results not only established

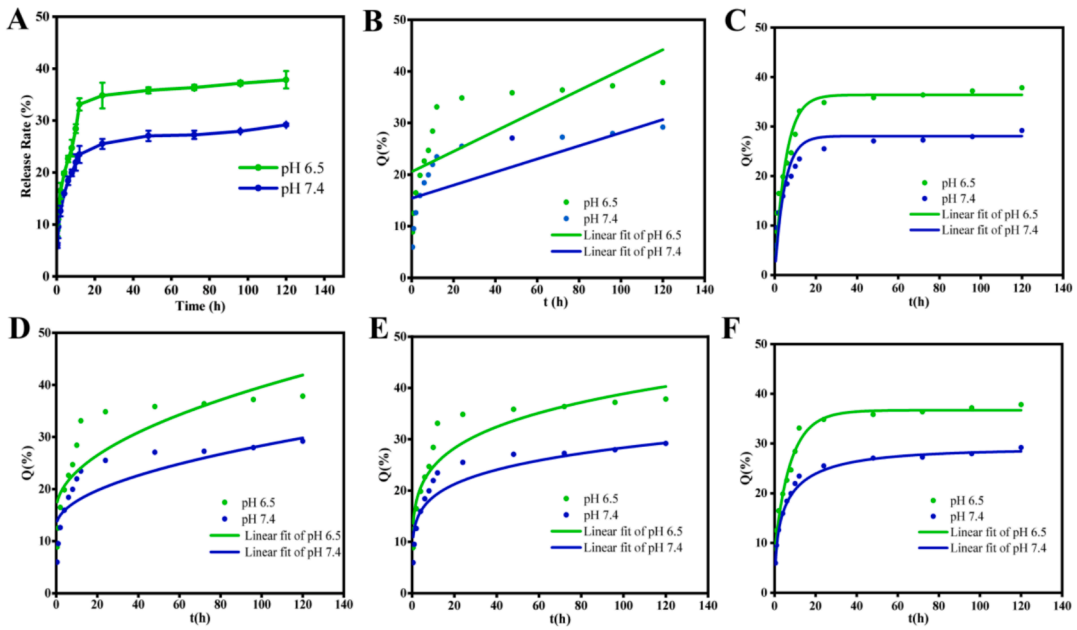


Fig. 4. (A) *In vitro* cumulative drug release for CDs/Ber at pH 6.5 and pH 7.4 (n = 3). The Ber release data at different buffers (pH 6.5 and 7.4) were fitted to (B) zero-order model, (C) first-order model, (D) Higuchi model, (E) Ritger-Peppas model, and (F) Weibull model of drug release. Experimental values were mentioned as spots, and predicted values were mentioned as lines.

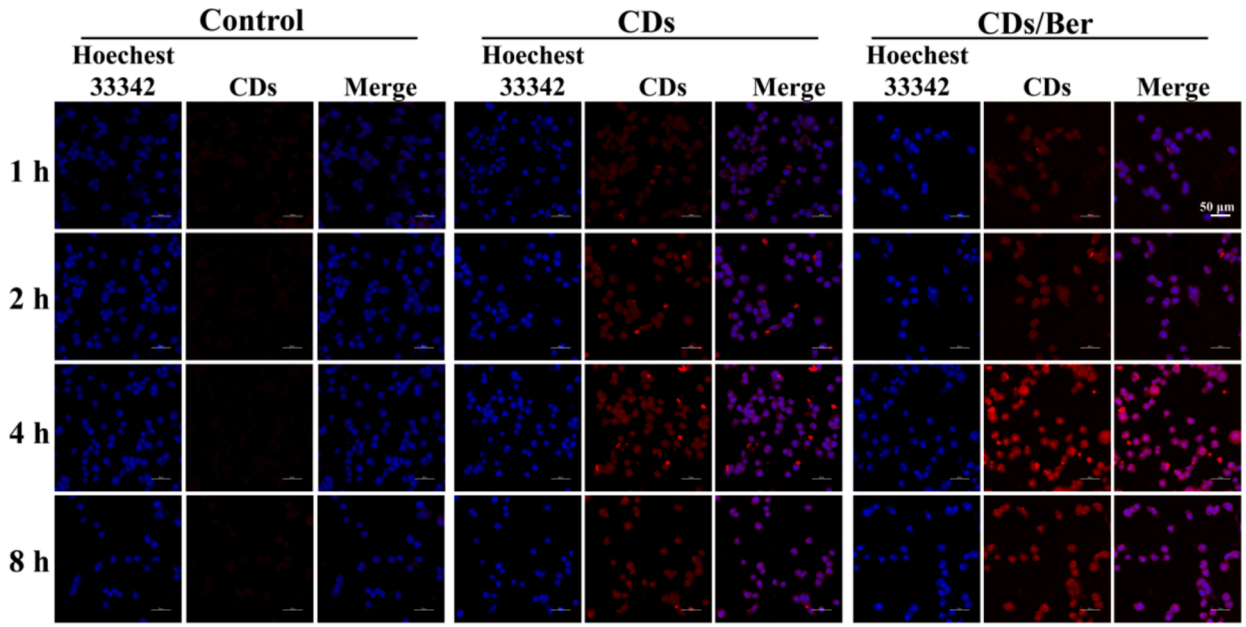


Fig. 5. *In vitro* cellular uptake of CDs and CDs/Ber with 200 μg·mL^{-1}. The red channels in the figure were the fluorescence of CDs and CDs/Ber. And the blue channel was the fluorescence of Hoechst 33342. Scale bar: 50 μm. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

the LLC cellular imaging of the CDs for cancer theranostics but also suggested the possibility of using the CDs to label cancer cells and locate drugs.

3.5. *In vitro* cytotoxicity

To estimate the biocompatibility *in vitro*, the CCK-8 assay was applied to investigate the cell viability of the prepared CDs and CDs/Ber against L929 and HPDE6-C7 cells. As shown in Fig. S4A and B, the cell viability of L929 and HPDE6-C7 cells in a series of concentrations (0–800 μg·mL^{-1}) of CDs were all above 80 %, and the result suggested that the CDs had excellent biocompatibility and security. To estimate the antitumor ability of the prepared CDs, Ber, and CDs/Ber *in vitro*, the CCK-8 assay was applied to investigate the therapeutic effects of CDs/Ber against LLC cells (Fig. 6A) and A549 cells (Fig. 6C). At a concentration of 10 mg·mL^{-1} under non-irradiated conditions, CDs and CDs/Ber displayed differential cytotoxic effects: LLC cell viability was suppressed to 60 %, in contrast to the 70 % viability retained by A549 cells, implying potential cell-line-specific defense mechanisms against nanoparticle-induced stress. Compared to normal cells (Fig. S4A and B), CDs and CDs/Ber exhibited significantly higher antitumor activity against both cancer cell lines, indicating their selective cytotoxicity

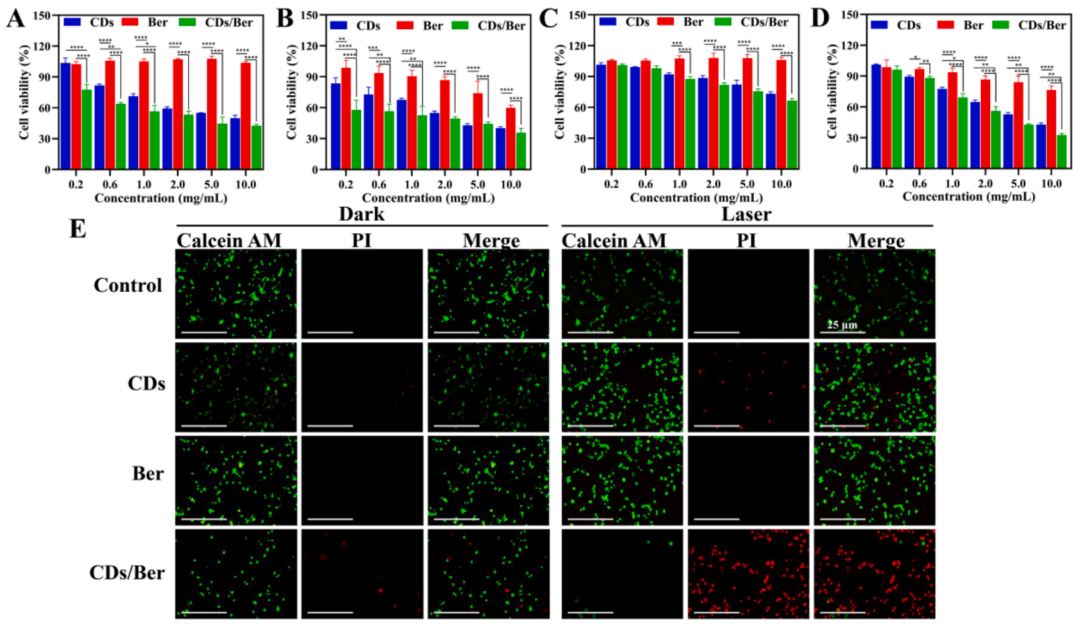


Fig. 6. Cell viability of LLC cells toward free Ber, CDs, and CDs/Ber incubation for 24 h (A) without NIR laser irradiation or (B) with NIR laser irradiation at a 2 W·cm^{-2} irradiation for 5 min (n = 5). Cell viability of A549 cells toward free Ber, CDs, and CDs/Ber incubation for 24 h (C) without NIR laser irradiation or (D) with NIR laser irradiation at a 2 W·cm^{-2} irradiation for 5 min (n = 5). (E) Calcein-AM/PI co-stained LLC cells with different treatments (Dark: without light irradiation; Laser: 808 nm laser irradiation for 5 min). Scale bar: 25 μm.

toward malignant cells. This cancer cell specificity suggests intrinsic tumor-targeting mechanisms, potentially mediated by dysregulated metabolic pathways or overexpressed surface receptors in cancer cells. For the NIR laser irradiation group, the cell viability of CDs and CDs/Ber was negatively concentration-dependent, proving that the generated heat that was converted from the NIR laser using CDs and CDs/Ber could kill tumor cells (Fig. 6C and D). Apart from this, the IC_{50} values for different groups are displayed in Table 1. For the Ber group, no IC_{50} could be determined before irradiation, whereas post-NIR irradiation IC_{50} values for LLC and A549 cells were 16.39 mg·mL^{-1} and 62.02 mg·mL^{-1}, respectively. However, both CDs and CDs/Ber groups showed reduced IC50 values after NIR irradiation compared to non-irradiated conditions. Notably, the CDs/Ber group exhibited significantly lower IC50 values of 1.374 mg·mL^{-1} and 3.451 mg·mL^{-1} in LLC and A549 cells, respectively. These results suggested that CDs/Ber could kill tumor cells quickly and CDs could be used as an ideal carrier in tumor treatment under NIR laser irradiation. In addition, the CI was used to evaluate the synergistic effect of chemotherapy and PTT: CI < 1 indicates synergism, CI > 1 indicates antagonism, and CI = 1 indicates additive effects. Although no IC50 value could be determined for Ber alone (due to insufficient cytotoxicity at tested concentrations), extrapolation via the combination index model revealed CI of CDs/Ber < 1. This sub-threshold efficacy strongly suggests a synergistic interaction between chemotherapy and photothermal therapy.

To observe intuitively the synergistic therapeutic effect of CD/Ber on LLC cells, the LLC cells were stained with calcineurin (AM)/propidium iodide (PI) to determine the live/dead cell ratio. In control groups, viable cells were consistently observed (as evidenced by green fluorescence staining) regardless of laser exposure (Fig. 6E), showing that the NIR laser itself did not cause damage to cells. There were almost no or only a few cell deaths in the group without laser irradiation. Cell death increased slightly after CDs treatment upon laser, which could be visualized by the red fluorescence, indicating that the CDs were effective at absorbing NIR light to generate sufficient heat levels capable of ablating cancer cells. It was worth noting that almost all cells in the CDs/Ber with laser died, which was much greater than the percentage of cells killed in the other treatment groups. These findings indicated that the combination of CDs and Ber actualized effective synergistic chemo-photothermal therapy, yielding a true "1 + 1 > 2" effect. The CDs/Ber had the potential for widespread application in cancer chemotherapy/PTT.

Table 1
IC_{50} value of CDs, Ber, and CDs/Ber groups.

	Formulations	IC_{50} (mg/mL)	
		Without NIR irradiation	NIR irradiation
LLC	CDs	9.733	4.095
	Ber	–	16.39
	CDs/Ber	3.112	1.374
A549	CDs	34.35	5.864
	Ber	–	62.02
	CDs/Ber	24.21	3.451

3.6. Biodistribution study

In cancer chemotherapy, the rapid clearance of chemotherapeutic drugs from solid tumors was one reason for treatment failure. Nanoparticles loaded with chemotherapeutic drugs prolonged the retention of drugs in the tumor through the subcutaneous injection [62]. The subcutaneous lung cancer xenograft model was selected for its technical feasibility and standardized assessment of therapeutic efficacy while retaining relevance to lung cancer biology. The biodistribution of CDs in LLC tumor-bearing mice was examined to evaluate the feasibility of fluorescent imaging *in vivo* and the tumor retention capability. We determined the fluorescence intensity of the tumor tissues after the subcutaneous injection of CDs and CDs/Ber. As shown in Fig. 7A, a strong fluorescence signal from the CDs/Ber was mostly observed in the tumor at 1 h and the accumulation of CDs/Ber was increased obviously with the extension of time. As demonstrated by real-time fluorescence monitoring in Fig. 7B, the cumulative amount reached the maximum at 24 h. This phenomenon may be associated with subcutaneous administration. The validation of intravenous injection will constitute the primary focus of subsequent research. *Ex vivo* fluorescence images of organs excised from mice (sacrificed at 24 h and 48 h post-injection) were also taken to observe the excretion pathway of the CDs/Ber. As

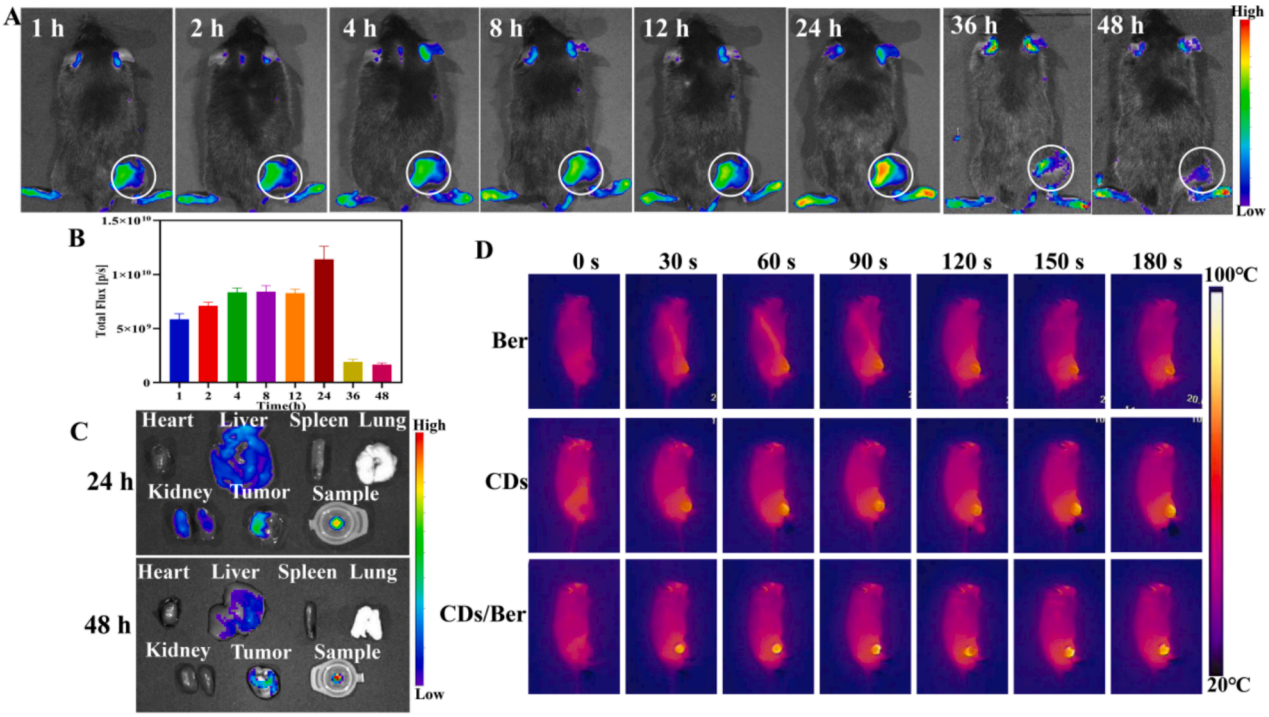


Fig. 7. (A) Fluorescence imaging distribution and (B) fluorescence intensity (the circle indicates the tumor site) of mice tumor-bearing mice injected with CDs/Ber (1 h, 2 h, 4 h, 8 h, 12 h, 24 h, 36 h and 48 h). (C) Fluorescence imaging of various organs and tumors of tumor-bearing mice 24 h and 48 h after injection of CDs/Ber. (D) Photothermal images in mice treated with different samples irradiated by NIR. NIR irradiation time was set at 0–180 s.

shown in Fig. 7C, the CDs/Ber accumulated more in tumors, followed by the liver. While the liver's fluorescence persisted throughout the 48-hour observation period, fluorescence became undetectable in renal compartments at this time point (Fig. 7C). This might be related to the metabolism of CDs/Be by the liver. These results indicated that CDs/Ber had excellent tumor imaging ability and targeted transport ability. In addition, we recorded temperature changes of the tumor through *in vivo* photothermal imaging of LLC tumor-bearing mice after injection with the saline, CDs, and CDs/Ber upon 808 nm laser irradiation (Fig. 7D). Following laser irradiation for 180 s, the temperature of the control group was 38.5℃ under irradiation at 0.8 $W{\cdot}cm^{-2}$. However, under the same irradiation conditions, the temperature of the tumor region injected with CDs and CDs/Ber increased to 59.8 ± 0.34℃ and 67.6 ± 1.09℃, respectively. When the temperature of the tumor was beyond 42℃ and sustained for several minutes, the persistent high temperature provided a more complete tumor ablation. Considering the small size (7 nm) and shape of the CDs/Ber, there was a possibility that they could penetrate the tumor cells by caveolae-mediated endocytosis alongside penetration from tumor areas or normal tissue blood vessels. Since caveolae avoided the lysosomal degradation pathway, the CDs/Ber taken via caveolae-mediated endocytosis were less likely to be degraded. These results revealed that the outstanding photothermal properties of CDs/Ber enabled the implementation of chemo-photothermal synergistic therapy for lung cancer.

3.7. *In vivo combined antitumor therapy*

The subcutaneous xenograft modelis particularly well-suited for evaluating therapeutic outcomesoflocalized photothermal therapy (PTT)andsystemic drug administration,as it enables real-time tumor volume monitoringandnon-invasive quantification of treatment responsesin a controlled microenvironment. Driven by the remarkable *in vitro* photothermal performance of CDs/Ber, we subsequently evaluated their therapeutic efficacy against tumors *in vivo*. Employing a one-time NIR irradiation to guarantee that normal cells were protected from high-temperature-induced cellular damage (Fig. 8A). During the whole treatment period, the mice's body weight was measured every other day. The research found that the mice in all groups did not exhibit abnormalities in feeding, drinking, or movement habits, and there were no significant differences in body weight (Fig. 8B). The results indicated that the mice displayed consistent behavioral patterns, and no significant toxicity was observed from these treatments. To evaluate the therapeutic effect, tumor sizes and weight were measured every other day. As shown in Fig. 8C, the tumor volume showed a continuous decrease in the CDs/Ber-treated mice group. After the 14-day observation period, tumors in all the mice were excised to visualize the therapeutic efficacy. Digital image of the excised tumors showed a substantial decrease in tumor volume in the CDs/Ber with laser-treated mice group contrary to the control and CDs mice groups, exhibiting an excellent PTT effect (Fig. 8D and E). Meanwhile, tumors in the Ber showed that the tumors were prominently ablated after the treatment. As shown in Fig. 8D, after NIR laser irradiation was performed 4 h post-injection, scars formed at the tumor site and fell off 14 days later, indicating that the tumor was completely ablated by PTT.

After 14 days of therapy, H&E histological analysis was carried out to assess the pathological sections of the tumor, as well as major organs in the mice such as the heart, liver, spleen, lung, and kidney. For all groups, no significant inflammation or damage was observed in the heart, liver, spleen, lung, and kidney compared to the control group (Fig. 8F), further confirming the excellent biocompatibility of the CDs/Ber. Additionally, the images of H&E-stained slices of tumors pronounced tumor-cell apoptosis for the CDs/Ber group compared with the saline and Ber mice groups. The treatment time of 14 days achieved with the CDs/Ber was similar to the time of 15 and 14 days required to treat tumors with gold nanoparticles [63] and carbon dots [35]. Moreover, the various intrinsic properties of CDs/Ber would enable their easy incorporation into other tumor treatment modalities such as immunotherapy, chemotherapy, and photodynamic therapy. Their excellent water dispersibility, biocompatibility, small size, facile synthesis, remarkable photothermal stability, and high photoconversion efficiency, along with

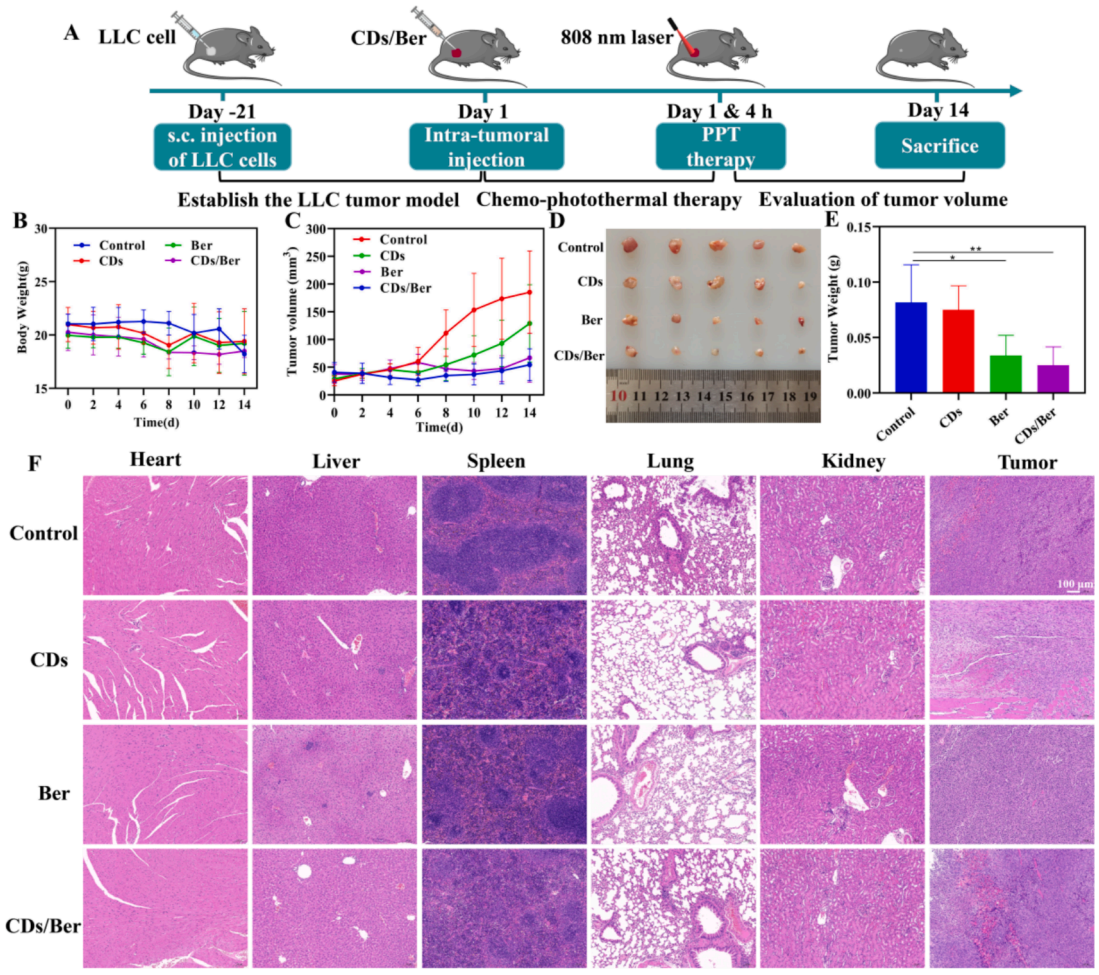


Fig. 8. (A) Schematic diagram of the construction and treatment process of the LLC tumor-bearing mice model. (B) Body weight, (C) tumor volume, (D) photographs of tumors, and (E) tumor weight of LLC tumor-bearing mice during various treatments (n = 5). (F) H&E staining images of various organs and tumor tissues after different treatments. Scale bar: 100 μm.

the advantages of single-dose administration and single NIR irradiation, made them suitable for integration with other drug carriers (such as nanoparticles, micelles, emulsions, and hydrogels, etc.) to enhance cancer PTT efficacy and complement other treatment modalities.

4. Conclusion

In summary, an innovative CDs-based multifunctional nanoplatform was engineered to overcome the inefficacy and systemic toxicity challenges inherent in conventional chemotherapy [64]. Distinguished from traditional nanocarriers and standard NIR-responsive photothermal agents, this system leverages its large specific surface area and honeycomb-like structure to serve dual therapeutic roles: (1) as a high-capacity drug carrier, ensuring efficient payload delivery, and (2) as a PTT agent, enabling synergistic chemo-photothermal treatment. This structural–functional integration significantly enhances therapeutic outcomes while mitigating toxicity risks associated with traditional approaches. Moreover, the CDs exhibited dual-modal fluorescence and photothermal imaging functionalities, enabling real-time spatial tracking and localization within biological systems. This dual-imaging capability positions them as a promising theranostic platform for precision-guided drug delivery and treatment monitoring. The CDs demonstrated sustained photostability and achieved a superior photoconversion efficiency of 54.9 % *in vitro*, outperforming gold nanorods [48], indocyanine green dyes [49], and carbon dots derived from alternative sources [48,50–53]. *In vivo*, the CDs/Ber system demonstrated a significant temperature elevation in the tumor region under NIR irradiation. The temperature increase caused by NIR irradiation further enhanced the efficacy of the drug, thereby achieving efficient synergy between chemotherapy and photothermal therapy. Furthermore, the CDs/Ber system exhibited low cytotoxicity toward normal cells, while the tumor cell mortality rate under synergistic therapy was significantly higher than that of chemotherapy or PTT alone. In tumor-bearing mice, the CDs/Ber system also demonstrated good biosafety and could mediate synergistic chemotherapy and PTT to effectively eradicate tumors without regrowth through a single-dose administration and single NIR irradiation. The developed nano-system held extensive prospects as an efficient collaborative platform for chemotherapy and PTT in antitumor treatments.

CRediT authorship contribution statement

Jiaxin Li: Writing – original draft, Methodology, Data curation. **Xiangxin Feng:** Writing – original draft, Methodology, Data curation. **Hongjun Jiang:** Methodology, Data curation, Conceptualization. **Xinhui Mo:** Methodology, Conceptualization. **Chang Liu:** Methodology, Conceptualization. **Xiongwei Liu:** Methodology, Conceptualization. **Tingting Feng:** Methodology, Conceptualization. **Ying Zhou:** Supervision, Resources, Funding acquisition.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Jiaxin Li reports financial support was provided by Science and Technology Program of Guizhou Province. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jcis.2025.137679.

Data availability

The authors do not have permission to share data.

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